Inside 6th-Generation Intel Core: New Microarchitecture Code-Named Skylake

Skylake's core, processor graphics, and system on chip were designed to meet a demanding set of requirements for various power-performance points. Its coherent fabric provides high-memory bandwidth from multiple memory sources, and its power management provides the largest dynamic power range among prior Intel processors. The Intel Architecture core delivers higher power efficiency, higher frequency, and a wider dynamic power range, supporting smaller form factors.

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Skylake—the code name for the project that implemented the 6th-generation Intel Core microarchitecture—delivers record levels of performance and battery life in many personal computing use cases and form factors. Intel Speed Shift Technology provides higher user-perceivable performance, responsiveness, and efficiency, with a greater benefit when the form factor is smaller and the power constraint is more restrictive.

Skylake's processor graphics deliver scalable performance, greater than 1 Tflops of computation, enhanced low-power media engines, flexible power management, and end-to-end 4K experience. The Skylake family of products lets platform developers choose from a wide range of platform capabilities, innovate with products for a wide range of thermal envelopes and I/O solutions, and optimize the system performance using the advanced Performance Monitoring Unit (PMU) capabilities. Skylake also introduces Intel Software Guard Extensions (Intel SGX), a new technology for application developers seeking to protect select code and data from disclosure or modification.

Fabric, Cache, and Memory Solutions

Skylake's fabric is a further development of the successful ring topology on-die interconnect first introduced in Sandy Bridge.[1] It is a scalable solution that can be configured for different derivatives of the same design. It supports variable numbers of Intel Architecture (IA) cores and multiple configurations of processor graphics. The fabric maintains memory coherency protocols between connected agents. It also has a built-in last-level cache (LLC). Skylake-coherent fabric was designed to provide high-memory bandwidth from multiple memory sources: the

Published by the IEEE Computer Society

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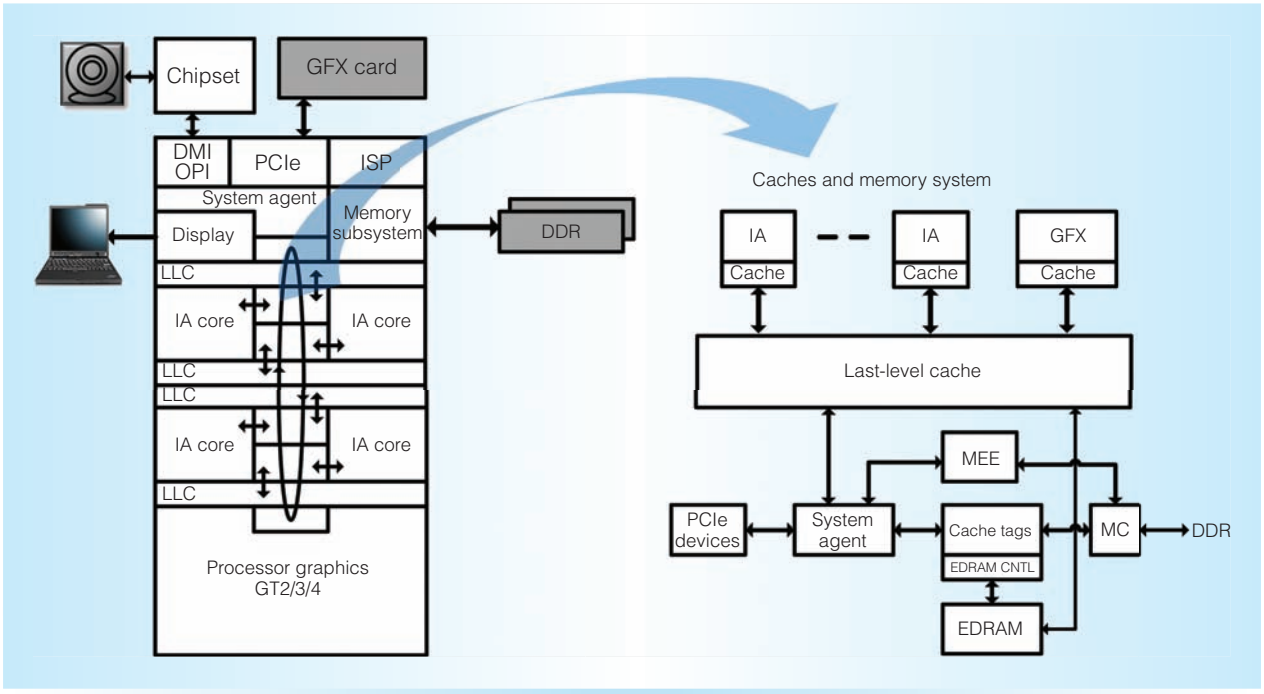


Figure 1. Skylake block diagram with zoom into cache and memory subsystem. The memory subsystem is shown with the eDRAM-based memory side cache.

LLC, embedded DRAM (eDRAM), and system memory.

The Skylake memory solution consists of a flexible and efficient system memory controller that provides the ability to use a Skylake system on a chip (SoC) in multiple platforms with different types of DDR technology. Some Skylake models that contain extended graphics capabilities include an eDRAM-based memory side cache. Figure 1 shows the SoC block diagram.

On-Die Interconnect

The Skylake fabric is defined to scale with the actual number of instantiated cores—that is, the total cache bandwidth is proportional to the core count. It can transfer full memory bandwidth of any memory configuration to the IA and graphics cores. The coherent fabric was optimized for better treatment of LLC misses. That lets us transfer more bandwidth from memory subsystems (eDRAM) at the same frequency or supply the same bandwidth as a prior-generation interconnect at lower frequency—that is, to save power. Currently, there is a selection of products with two or four cores, with or without eDRAM, and three different sizes (and thus consumed bandwidth) of processor graphics. The Skylake fabric can deliver approximately 100 Gbytes per second (GBps) LLC bandwidth (depending on the frequency) to every core simultaneously. From eDRAM, it can deliver about 114 GBps (57 read and 57 write) to the processor graphics.

Memory Subsystem

The Skylake memory subsystem includes an external memory controller, memory encryption engine, and memory-side cache based on eDRAM.

The memory controller supports DDR3, DDR4, and LPDDR3. It was extended to support 2,133 MHz for DDR4. It is optimized for the best utilization of DDR bandwidth capabilities together with guaranteed quality of service for isochronous requests from an integrated display engine and integrated image processing unit.

Memory encryption capability is an essential part of Intel's Software Guard Extensions (SGX) technology.[2,3] This block is part of the memory subsystem. Designated memory requests are routed to the memory encryption

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engine, which provides data encryption and integrity functionality before or after reaching the system memory for writes and reads, accordingly (see the memory subsystem diagram in Figure 1).

A significant change in Skylake's memory hierarchy solution is the introduction of the eDRAM-based memory-side cache. In previous generations, eDRAM was used as an L4 cache and was available only for IA and graphics requests.[4] As a result, it was architecturally visible in the way that instructions like CLFLUSH or WBINVD saw it as part of the processor cache. This means that software was responsible for flushing the modified data from the cache before it could be accessed by nonsnoopable device requests. With memory-side cache, every device or processor core request looks up eDRAM, which guarantees that all agents see the same data. In the case of a miss, the accessed address may be allocated in eDRAM based on CPU settings, but in case of a hit, the most updated data will be forwarded to the requestor. For optimal performance, the graphics driver can choose caching of particular data in the eDRAM only and not in the LLC. There are now two products in the Skylake family that use the eDRAM solution. One has four cores with GT4 graphics and eDRAM (128 Mbytes), and the second has two cores with GT3 graphics and eDRAM (64 Mbytes).

Power Management

Skylake power management was architected and designed to provide the largest dynamic power range among prior Intel processors—4.5 W in passive cooled small systems and up to 95 W in high-performance desktop computers.

Skylake introduced a revolutionary approach to power management called Intel Speed Shift technology. Operating systems traditionally have been responsible for managing performance and energy by controlling the CPU voltage and frequency via P-states. When the processor utilization is high, the operating system (OS) assumes that high performance is needed and commands the CPU to run at a high-performance P-state. When utilization is low, the OS commands the CPU to run at a lower-performance, more energy-efficient P-state, saving energy with no impact on performance. The OS, however, has two fundamental limitations when it comes to making those optimizations:

- The processor utilization evaluation is performed over a few tens of milliseconds, because anything faster than that would have been too intrusive.
- The OS has limited visibility of the workload's instantaneous runtime behavior and microarchitectural characteristics. For example, the OS does not know if the workload is bound by memory latency. Running at a high clock frequency while waiting for data to return wastes energy, with limited gain in performance.

With Intel Speed Shift technology in Skylake, the CPU assumes full responsibility of power, performance, and energy efficiency, not the OS. A fully autonomous control algorithm replaces the OS's P-state selection and achieves up to 14 percent improvement over the previous generation in performance and energy efficiency of user-visible scenario benchmarks.

Intel Speed Shift Technology Architecture

The traditional power management interface consists of enumeration and control. The enumeration for legacy OSs comprised the following:

- P1: guaranteed frequency that can be achieved at worst-case conditions.
- P*n*: the most energy-efficient frequency that corresponds to the minimum operation voltage.
- P0: opportunistic turbo frequency (Turbo boost technology) that can be used in cases such as lighter load applications and single core.

The OS used core utilization to determine and control the requested P-state over the P1 to P*n* range; when more performance was needed, the OS requested turbo frequency, in which actual control was transferred to the processor. With the new Speed Shift interface, Skylake lets the OS define the minimum quality of service (QoS), maximum allowed frequency, and energy performance preferences. An additional

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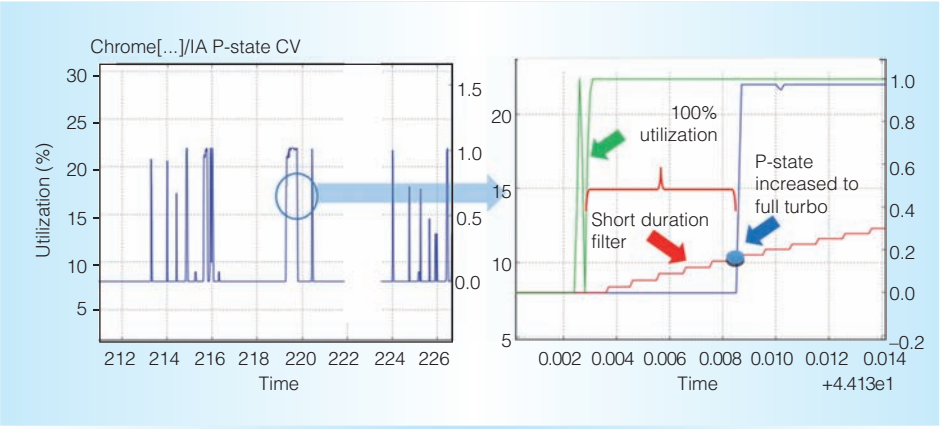


Figure 2. Responsiveness algorithm. The figure identifies user action by observing the increase in utilization after a long idle time and jumping to maximum frequency.

explicit P-state request is also provided for legacy operation.[5]

Intel Speed Shift Technology Algorithms

P-state selection is based on utilization, as before, but now there is better observability to the workloads' runtime characteristics at short time intervals with the following improvements.

First, thread dependencies of consumer producer workloads was improved by thread dependencies observability. In consumer-producer scenarios, the CPU might seem underutilized while in fact threads are waiting for another thread to complete: thus, lowering the P-state could result in prolonging the producer thread. Second, user responsiveness was improved. Interactive workloads are characterized by high CPU utilization bursts separated by idle time intervals. Some frame-based workloads also demonstrate heavy load at one or a few frame intervals. Figure 2 describes the Skylake algorithm. The Speed Shift technology algorithm detects fast utilization changes at 1-ms intervals after a long idle period. An interval counter filters out short-duration activities, such as interrupt service routines. Crossing the counter threshold indicates user responsiveness demand, and P-state then kicks in to full turbo. Measurements on the Skylake system demonstrate 14 percent performance improvements on user-interactive workloads represented by WebXPRT. This algorithm detects and responds to user action an order of magnitude faster than the OS's demand-based algorithm.

Energy-Efficiency Control

Often a user will want to trade performance for lower energy consumption to achieve longer battery life, reduced fan noise, or lower skin temperature. User preference is managed by the OS interfaces that define several policies, such as *balanced* and *energy efficient*. Figure 3 describes the energy efficiency as a function of frequency.

The energy consumption as a function of frequency has a variable global minimum $P_{\text{Most Efficient}}$ (Pe). Firmware calculates this point every millisecond using an algorithm called Energy Aware Race to Halt (EARtH).[6] An energy-optimization algorithm uses the P-state that is calculated using the demand-based algorithm described earlier and performs energy optimization in two distinct regions.

If the utilization is low and there's no need to run the processor at a high-performance state, the lowest (performance) possible P-state will be selected. Obviously, running lower than Pe is suboptimal, because it results in increased energy consumption. If the Autonomous algorithm or the OS requests a P-state lower than Pe, the frequency is clamped to Pe and follows Pe as it changes with the workload characteristics. Only physical constraints such as high temperature can force the frequency lower than Pe.

If performance is needed, P-state is increased at the cost of energy efficiency. The user or OS can provide a hint as to how

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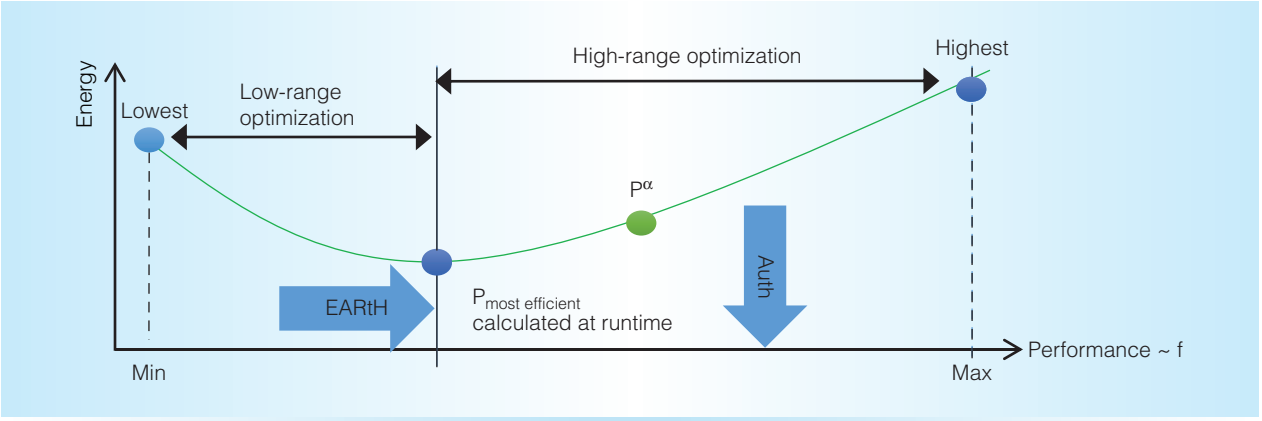


Figure 3. Energy as a function of frequency. Race to halt increases processor energy while saving platform and chip energy. A frequency that minimizes overall energy point exists.

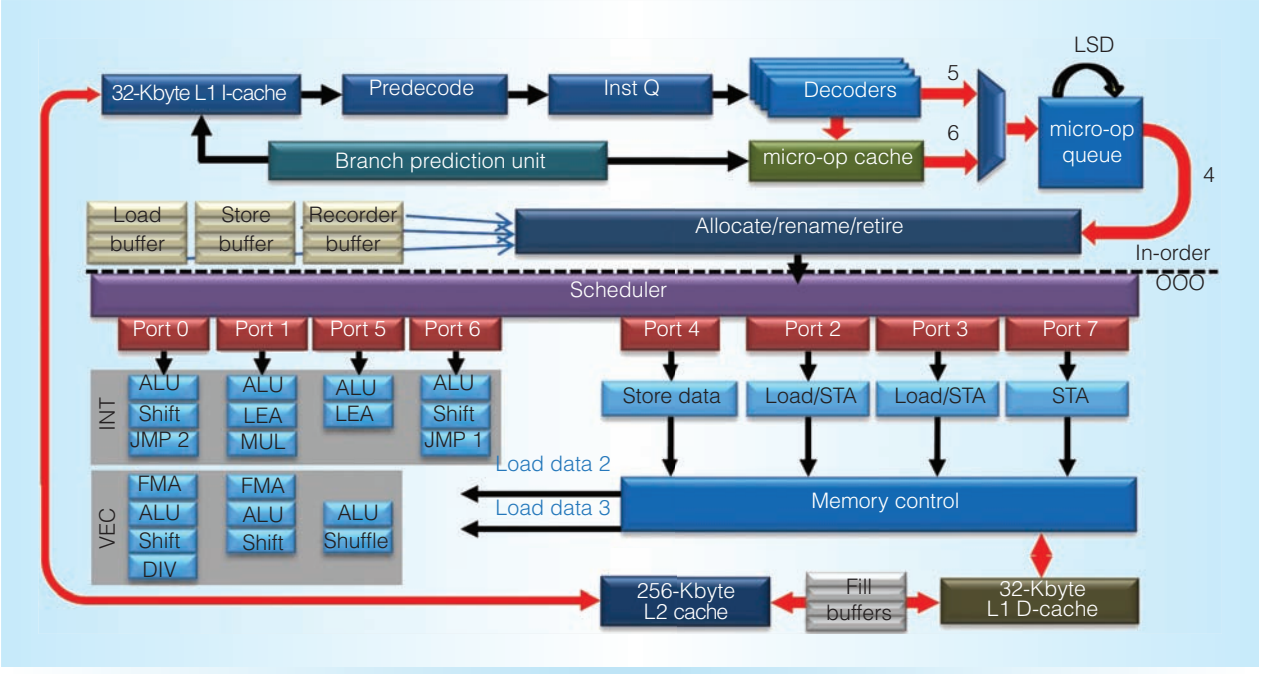


Figure 4. Skylake core block diagram.

much energy is to be spent for a given amount of performance. Skylake picked the well-known *Energy * Delay*$^{\alpha}$, or ED$^{\alpha}$, for optimization.[6] The OS provides α, and the power management algorithms continuously monitor the workload runtime characteristics to meet or exceed ED$^{\alpha}$. This not only saves power but also saves the scarce Turbo budget to boost frequency above nominal frequency for computation-bound phases.

Core Microarchitecture

The Skylake Core extends the Broadwell microarchitecture to deliver higher instructions per cycle, higher frequency, and a wider dynamic power range. The core becomes wider and deeper in a power-efficient way, bringing more performance into lower-thermal-power envelopes. Figure 4 shows the core block diagram, including the front end, out-of-order structures, execution units, and memory subsystem.

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Skylake adds multiple improvements in the core's front end for supporting higher-bandwidth instruction supply, reduced branch misprediction penalties, and reduced instruction-cache miss penalties. Skylake's branch predictor accuracy is improved, and its capacity is increased. Furthermore, the penalty for mispredicted direct jump targets is reduced. Skylake supports a more aggressive instruction prefetch that reduces the penalty of instruction-cache misses. Both the legacy decode pipeline bandwidth and the micro-op cache bandwidth are increased: the legacy decode pipeline peak bandwidth is increased from 4 to 5 micro-ops per cycle, and the micro-op cache peak bandwidth is increased from 4 to 6 micro-ops per cycle. Skylake improves LSD (Loop Stream Detector) mode, which plays loops with a small loop body directly from the micro-op queue, thus turning off the rest of the front end's clocks. By increasing the size of the micro-op queue from 28 to 64 micro-ops per thread, Skylake increases the LSD coverage, saving more power.

Skylake's out-of-order engine buffers are deeper, allowing the extraction of more instruction-level parallelism. Skylake allocates 4 micro-ops per cycle into the instruction scheduler (which is in charge of dispatching micro-ops to execution units [EUs]) and into the reorder buffer (which is in charge of fault handling and retirement). The instruction scheduler is increased from 60 to 97 micro-ops, and the reorder buffer size increases from 192 to 224 micro-ops. The instruction scheduler entries are activated on demand, by means of clock gating, thus saving power when not used. The integer register file increases from 168 to 180 entries.

Skylake improves the bandwidth and/or reduces the latency of many of the arithmetic EUs. The throughput of division operations increases by 1.67 to 2.33 times, depending on the data type and size. The latency of most floating-point division and SQRT operations is reduced. The throughput and latency of floating-point ADD, MUL, and FMA are balanced and made more uniform. The latency for AES instructions is reduced from 7 to 4 cycles.

Skylake's core memory subsystem improves the performance of load and store operations. For load operations, Skylake reduces store-to-load forwarding latency and significantly reduces page-split load penalty. For stores, Skylake uses deeper store buffers to increase store throughput and issues requests to the L2 cache for stores that miss in the L1 cache earlier, thus reducing the store miss penalty. Skylake also increases the bandwidth from the in-core L2 cache to the shared L3 cache, allowing higher streaming stores bandwidth. Skylake supports a new instruction to flush cache lines, CLFLUSHOPT,[2] which can be used, for example, for efficiently flushing a page out of the cache, up to nine times faster compared to the legacy CLFLUSH instruction.

Skylake employs a new in-core power virus protector mechanism that reduces the power virus Cdyn by preventing typical power virus scenarios (high density of high-power instructions). This reduces the required voltage margin, which reduces the core's power by approximately 3 percent and improves the core's power and performance. It also improves SMT mode burst performance by up to 11 percent.

Performance Monitoring Unit

Skylake made significant enhancements to the PMU that ultimately enable software developers to squeeze out the best performance. Table 1 summarizes the key enhancements.

A new Architectural PerfMon version 4 is introduced with better virtualization, atomicity (freezing) interface, and sharing of the PMU resources.[7] Skylake offers a shorter yet higher-quality PMU event list that improves support for the Top-down Microarchitecture Analysis Method—a structured drill-down process to identify performance bottlenecks[8]—as an example usage. The Precise Event-Based Sampling (PEBS) as well as the Last Branch Record (LBR) facilities were also enhanced. Equipping LBR with time is a key enabler for a new class of usages. Our previous work has more information on these enhancements and the enabled usages.[7]

Processor Graphics

Skylake's Gen9 graphics provides many new features that are designed to maximize energy efficiency and bring out the best visual

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Table 1. Comparison of Skylake's performance monitoring unit versus the predecessor.

Feature		Haswell	Skylake
PerfMon	Arch PerfMon version	3	4
	Events coverage (of microarchitecture)		Richer
	Events quality		Better
	Top-down Microarchitecture Analysis	Basic (SMT off)	Accurate (SMT on)
PEBS	Front-end events coverage	No	Yes
	Time-stamp counter included	No	Yes
LBR	Timing information	No	Yes
	Number of entries	16	32

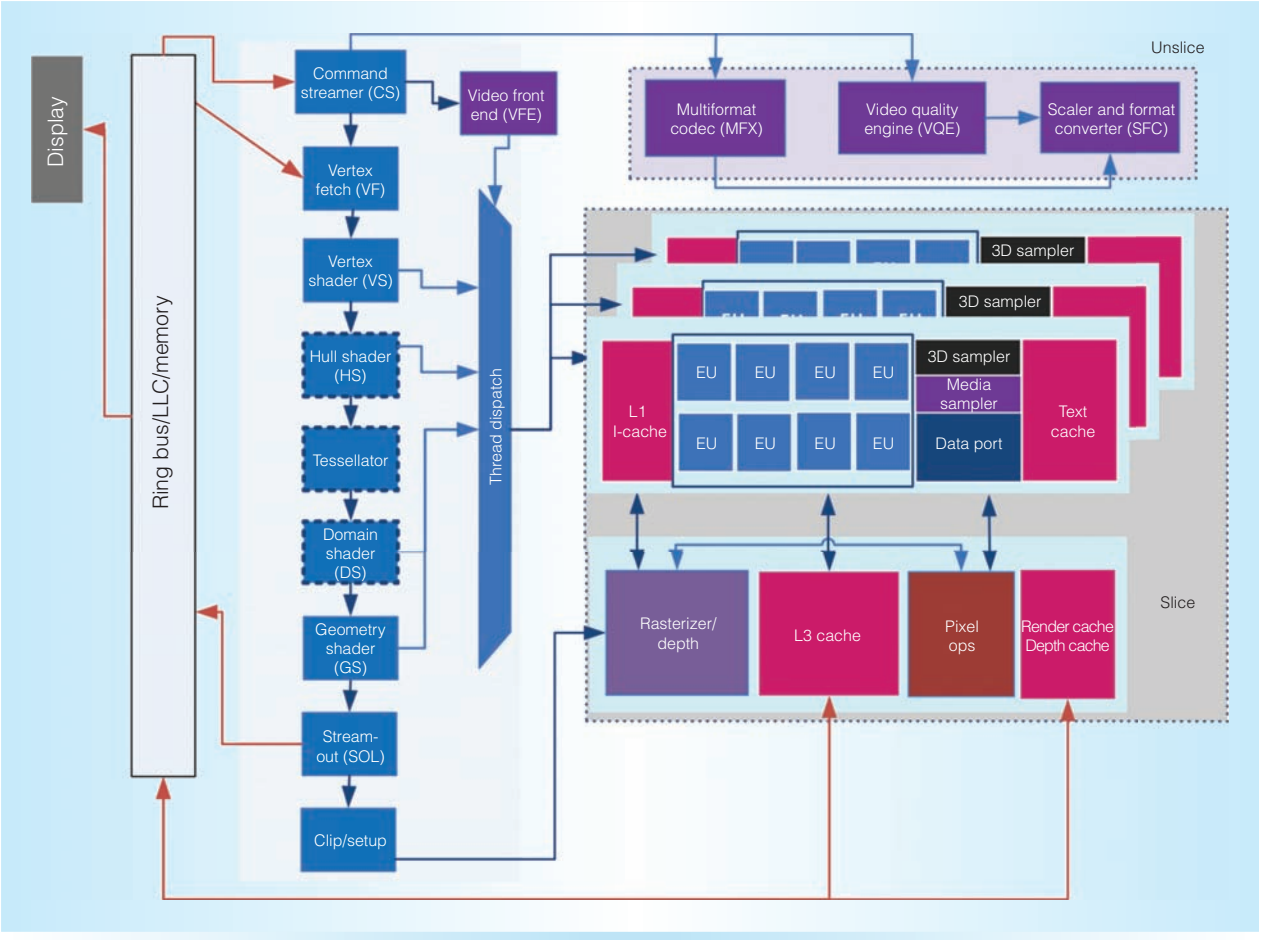


Figure 5. Gen9 processor graphics block diagram. A GT2 SKU with fixed functions on the Unslice and shared functions and 24 execution units (EUs) for rendering and computing on the Slice.

experience for gaming and media. To understand how power management features on Skylake achieve better energy efficiency, we first take a high-level view of the graphics architecture (see Figure 5). The architecture is partitioned into two major domains: Unslice and Slice. Fixed functions are in the Unslice. The render, computation, and programmable media functions are on the Slice. Each Slice consists of three subslices and a

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slice common. A subslice has eight EUs, a 3D and media sampler, and first-level caches. Additional L3 cache and pixel operations are in the slice common. The HD Graphics SKU has one slice, and Iris and Iris Pro Graphics have two and three slices, respectively, with additional eDRAM.

The scalability of 24 to 72 EUs can easily provide more than 1 Tflops of 32-bit computation power (2 Tflops 16-bit) that operates within 45 W of thermal design power. Graphics rendering pipeline throughput has been increased to better support demanding 3D rendering. Lossless render compression further reduces the memory bandwidth and can reduce the power consumption.

Several media engines have been rearchitected to better support low-power media. The Multi-Format Codec engine not only adds new codec features, such as H.265/HEVC (High Efficiency Video Coding) decode and encode, but also includes a dedicated low-power, H.264/AVC (Advanced Video Coding) low-latency encoder called Quick Sync Video FF (Fixed Function) mode, which supports real-time applications without using the main render pipeline. This is particularly useful for wireless display, video chat, or camera recording. The Video Quality Engine has improved power efficiency to deliver high-quality video processing features at a small power budget. In addition, a brand-new engine called the Scaler and Format Converter is added to the Unslice for image resizing and format conversion that can be used during video playback and other scenarios without using the render pipeline, too. Media operations also employ lossless memory compression to reduce intermedia engine traffic for better performance and power.

Graphics power management has increased flexibility over previous generations. Among many power management improvements, we highlight several important changes that directly provide performance benefit.

- *Slice/Unslice decoupling*. Slice and Unslice can now operate in different clock domains and can be power gated independently. The main render engine in the Slice can be completely powered off while operating media fixed function in the Unslice. Therefore, Skylake can achieve whole-day Ultra-HD 4K video playback on a typical mobile PC.
- *Finer granularity power management*. Each slice or pair of EUs can also be turned on and off to balance the performance and power. Each subslice can be clock gated if it is not actively used. Programmable encoding is a use case that can benefit this feature. It uses the programmable media samplers and EUs to complete the motion search, decision making, and rate control. In high-quality encode mode, which is computationally intensive, we can turn on all slices. In high-speed encode mode, which has limited parallelism, we can keep one slice on and even turn off unused EUs for better performance and power efficiency.
- *Multiple Voltage-Frequency (V-F) curves and C-state selection*. Because the GPU can operate in different configurations, defined by how many slices are turned on and off, the ideal V-F curve changes for each one. Skylake supports multiple curves, allowing higher maximum Graphics Turbo frequency when the GPU is in a smaller configuration, enabling increased performance. Furthermore, the algorithm to select which C-state to enter and exit is more aggressive and can quickly recognize the machine's active and idle state and switch the GPU to its optimal state.

Performance Results

Skylake delivers significant CPU and 3D gaming performance improvements over the prior-generation architecture, Broadwell, in the 4.5 and 15 W power envelopes (see Table 2). For the first time, Intel Core U processors (15 and 28 W) with Iris Graphics include eDRAM, which can boost 3D gaming performance by 40 percent in bandwidth-constrained scenarios.

At the other end of the spectrum, power reductions during video playback and web browsing enable Skylake to deliver up to 1

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Table 2. Performance and battery life of Skylake versus its predecessors.

Skylake processor performance vs. its predecessors	Thermal design power (TDP)	Computational performance	3D gaming performance	Battery life
6th-generation Intel Core m7 processors, mobile Y-processor, Intel HD graphics	4.5 W	Up to 1.25 times vs. Broadwell	Up to 1.45 times vs. Broadwell	Up to an additional 1.2 hours vs. Broadwell
6th-generation Intel Core i7 processors, mobile U-processor, Intel Iris graphics	15 W	Up to 1.28 times vs. Broadwell	Up to 1.71 times vs. Broadwell	Up to an additional 1.1 hours vs. Broadwell
6th-generation Intel Core processor, socket 1151, low power	35 W	Up to 1.51 times vs. Haswell	Up to 1.26 times vs. Haswell	Not applicable on desktop
6th-generation Intel Core processor, socket 1151, standard power	65 W	Up to 1.36 times vs. Haswell	Up to 1.30 times vs. Haswell	Not applicable on desktop
6th-generation Intel Core processor, socket 1151, 91 W	91 W	Up to 1.30 times vs. Haswell	Up to 1.26 times vs. Haswell	Not applicable on desktop

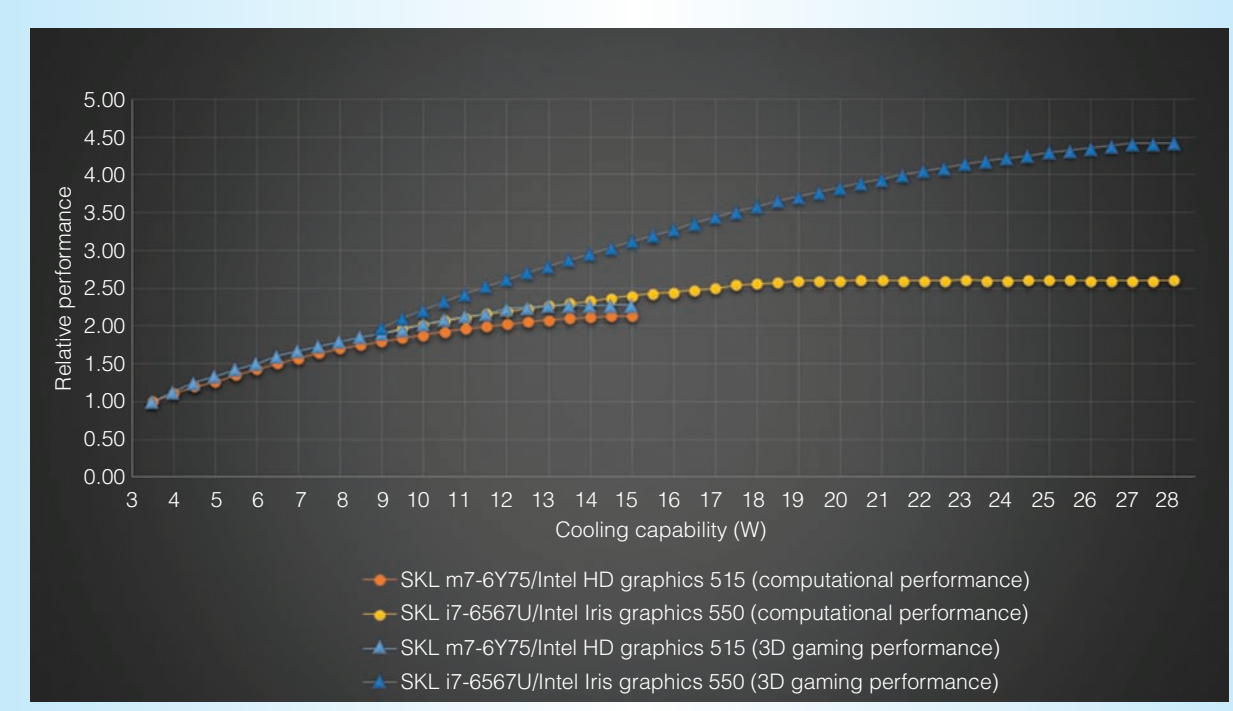


Figure 6. Skylake mobile platform performances scale with cooling capability, up to four times the performance from 3.5 to 28 W.

hour of additional battery life over Broadwell, assuming all other platform components (such as panel, storage, and memory) are the same.

Figure 6 shows that Skylake processors aimed at fan-less 2-in-1 form factors and actively cooled Ultrabooks are the most scalable Intel processors yet. Computational

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performance can scale 2.6 times, and 3D gaming performance can scale 4.4 times from 3.5 W to 28 W of cooling capability. The additional CPU cores and higher turbo frequency on the 91-W Skylake desktop processor can further extend the computational performance to six times. This wide dynamic range of performance capability with a single SoC allows original equipment manufacturers to pick a target performance level and design the thinnest form factor that can deliver it.

Skylake delivers record levels of performance and battery life in many personal computing use cases and form factors. Intel Speed Shift Technology provides higher user-perceivable performance, responsiveness, and efficiency. The Skylake family of products allows platform developers to choose from a wide range of platform capabilities, innovate with products for a wide range of thermal envelopes and I/O solutions, and optimize the system performance using the advanced PMU capabilities. Skylake also introduces Intel Software Guard Extensions, which assist application developers who are seeking to protect select code and data from disclosure or modification. MICRO

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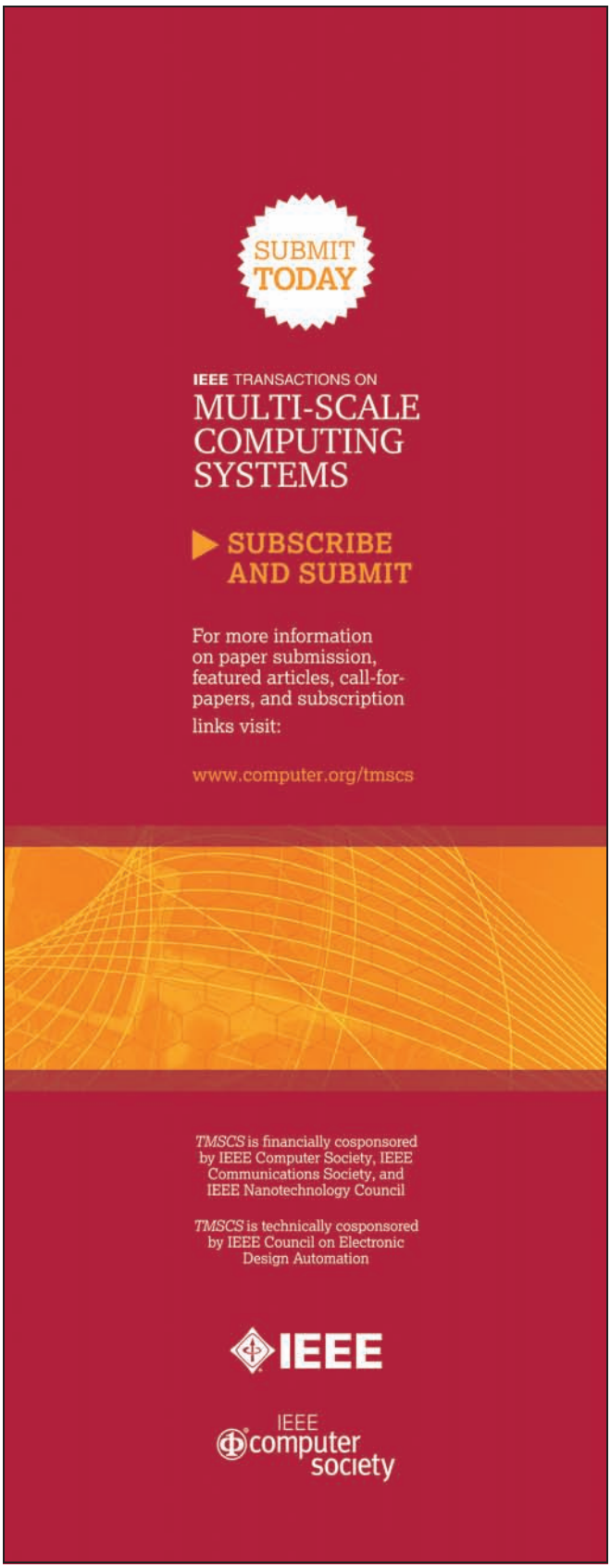
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